Lab 9 Worksheet — SOHO Network Configuration — DHCP, DNS, NAT and Port Forwarding

Name: ____ **Date:** __

Exam objective: Configure a SOHO router including DHCP scope, DNS, NAT, DHCP reservations and port forwarding, and verify each service from a client (Core 1 objective 2.5).

Record as you go

Step	What you did	What you observed
1	Plan the addressing before touching the router: choose 192.168.50.0/24, reserve .1 for the gateway, .2 to .99 for static assignments and .100 to .250 for the DHCP scope.	
2	Verify the plan in IP Calculator by entering 192.168.50.0/24 and confirming the usable host range covers every allocation you made.	
3	Configure the DHCP scope on the router with the start and end addresses from your plan, and set the lease time.	
4	Define the exclusion range covering .2 to .99 so the DHCP server never hands out an address reserved for static assignment.	
5	Create a DHCP reservation binding a printer's MAC address to a fixed address such as 192.168.50.20, so it always receives the same IP.	
6	Set the DNS servers the router hands to clients, and record whether you used the ISP resolver or a public one such as 8.8.8.8 or 1.1.1.1.	
7	On a client, release and renew the DHCP lease and record the address, mask, gateway and DNS servers received.	

Step	What you did	What you observed
8	On the Killercoda playground, inspect the equivalent client-side configuration and identify the interface address, mask and default route.	
9	Verify DNS resolution works through the configured resolver and record the answer section of the response.	
10	Explain how NAT lets many private hosts share one public address, and identify from the router status page what your public address is.	
11	Create a port-forwarding rule directing external TCP 3389 to an internal host, and state the security risk of exposing RDP directly to the internet.	
12	Document the complete configuration in a table a colleague could use to rebuild the router from scratch after a factory reset.	

Verification

Success criterion: The client receives an address inside the DHCP scope and outside the exclusion range, the reservation returns the same address after a release and renew, DNS resolves successfully through the configured resolver, and the port-forwarding rule is documented with its security risk stated.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?